第十章 文件系统接口

1. Consider a file system where a file can be deleted and its disk space reclaimed while links to that file still exist. What problems may occur if a new file is created in the same storage area or with the same absolute path name? How can these problems be avoided?

Problems: ① Stale reference: Old links may access the new data incorrectly.

② Name aliasing confusion: If a new file is created with the same pathname, processes holding old links may do a wrong dereference.

How to avoid: ① Reference counting on inodes

② Defer deletion

③ Use unique file IDs

2. The open-file table is used to maintain information about files that are currently open. Should the operating system maintain a separate table for each user or just maintain one table that contains references to files that are being accessed by all users at the current time? If the same file is being accessed by two different programs or users, should there be separate entries in the open file table?

(1) It uses both of them, for sharing and mapping.

(2) Yes, because each access may have its own file offset, mode and flags.

3. What are the advantages and disadvantages of a system providing mandatory locks instead of providing advisory locks whose usage is left to the users' discretion?

Advantages: ① Strong correctness guarantee, since processes can't ignore locks.

② Prevents accidental corruption by buggy or careless programs.

③ Good fore high-integrity shared data

Disadvantages: ① Less flexible

② Can degrade performance or cause more blocking.

③ Greater risk of deadlock.

4. Discuss the merits and demerits of supporting links to files that cross mount points (that is, the file link refers to a file that is stored in a different volume).

Merits: ① Convenient and transparent: Users can link across volumes as if it's one namespace.

② Better organized: Data can live where it fits without breaking directory.

③ Space balancing: Move files between volumes without changing the tree.

Demerits: ① More complexity: Harder for path resolution.

② Failure: If the target volume is unmounted or offline, links will become invalid.

③ Consistency and security: May cause some security problems.